

Today's topics

Revise page navigation n session tracking with diagram n code walk through

How to write DB independent (partially at least !) web application ?

Page navigation with server pull

Servlet life cycle with thread pool (executor framework)

Web app listener

Enter JSP (If time permits)

1. How to make web app DB independent (partially)?

1.1 Parameterize the Servlet

(parameters --db URL , user name , pwd)

Using Parameterized ctor - Failed !!!!!!!

--will NOT work !(since WC will call def arg less ctor ONLY!!!!)

Solution --use javax.servlet.ServletConfig --i/f

Represents servlet specific config object.(one per servlet)

Who creates it ? WC (implicitly)

When ? After creating servlet instance (using def ctor) n before calling the init method of the servlet.

For which purpose ? to supply servlet specific init params

How to provide init params ?

1. Using annotation

OR

2. Using XML tags --- preferred approach !

(since it does not involve changing Java code !!!)

In web.xml --

<servlet>

<servlet-name>

<servlet-class>

<init-param>

<param-name>

<param-value>

</...>

<load-on-startup>

</servlet>

How to access these init params from the servlet ?

ServletConfig API

public String getInitParameter(String paramName)

Steps

1.1 web.xml

add 3 init params in a servlet

(db config : db URL, user name ,password)

URL : jdbc:mysql://localhost:3306/dbName

1.2 In Login Servlet

Read init params : from ServletConfig

1.3 pass these params to DBUtils
openConnection : to actually open a new connection.

2. Page Navigation with server pull.

New Scenario

Student Admission Form (Accept student details : first name , last name , entrance test score,course,admission status)

--> submit this info to Student Admission Servlet. Validate if student has min score for a specific course .
Accordingly give admission or deny admission.

Navigate the client to result page n display suitable message.
(refer to flow diagram)

Request Dispatching technique (Server Pull)

refer : readme n diagrams

Student Admission

Student POJO

firstName,lastName, --string

course : enum

score : int

admissionStatus : boolean

2.1RD's forward scenario

2.2 RD's include scenario

3. Pending -

Complete Servlet Life cycle (including thread pool)

Executor Framework (used by WC to support concurrent handling of multiple client requests)

CGI Vs Servlets :(reading H.W)

4. Web app listeners

Add a listener in the existing web app n test it.